curAHack 2026

Challenge 5

Gut Microbiome-Based CVD Risk Prediction

Advancing CVD Prediction using Machine Learning

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Building on the Gut Age Index (GAI) Pipeline

Bao Z. et al. (2024), Scientific Reports 14:31143

Motivation

17.9M

CVD deaths per year globally

68% Acc

CVD prediction based on Gut Microbiome shows promise

Current microbiome-based CVD prediction has critical gaps:

- Sparse input data is hard to process by classical ML algorithms
- Models fail to generalize across populations
- Severe class imbalance biases models toward majority class
- ➢ Performance is sub-optimal and inappropiate for clinical use

YOUR MISSION

Build a better CVD risk prediction system from gut microbiome data

Target: Balanced Accuracy

> 70%

Target: AUC (ROC)

> 0.75

Starting point: GAI paper achieves 68% balanced accuracy on AGP CVD data.

The GAI Pipeline: Your Foundation

Data Acquisition

- ~6,000 tabular samples per cohort (AGP, GGMP)
- Aggregated Healthy/Non-Healthy Information based on disease data

Available Code

- Code for training and evaluating models is already present
- Strong foundation for building new models

Age Regression

- Trained CatBoost/LightGBM models predict chronological age from OTU abundances
- GAI = predicted age – actualage provides a supplementary feature

Hackathon Timeline (Example)

Day 1 **Session Part 1 (2h)**	Understand baseline GAI pipeline, download baseline codes
Day 1 **Session Part 2.1 (1h)**	Reproduce paper's baseline GAI
Day 1 **Session Part 2.2 (2h)**	Feature engineering
Day 1 **Session Part 2.3 (3h)**	Train improved models
Day 2 **Session Part 3 (3h)**	Handle imbalance, cross-cohort test, finalize results
Day 2 **Final Session (1h)**	Prepare presentation

Where Your Improvements Come From

Area	Paper's Approach	Your Opportunity
Features	Raw OTU counts	**diversity metrics, microbial ratios, OTU pair graphs, feature selection, etc**
Compositionality	Not addressed	**Centered Log-Ratio (CLR) transformation**
Class imbalance	Balanced accuracy only	**SMOTE, class weights, stratified cross-validation**
Model approach	Indirect (age → threshold)	**Hybrid: GAI, direct classification**
Tuning	50 iteration random search	**Bayesian optimization (e.g., Optuna)**
Interpretability	Feature importance	**SHAP, biological validation**

Common Pitfalls to Avoid

✕ Data Leakage

Compute CLR, SMOTE, normalization stats only on training data. **Never include test data in preprocessing.**

✕ Overfitting to Small CVD Class

180 samples can be memorized. **Use cross-validation, not single train/test split.**

✕ Raw Accuracy Metric

With 10:1 imbalance, 91% accuracy = always predict healthy. **Use balanced accuracy.**

✕ Skipping Interpretation

A black-box model is scientifically uninteresting. SHAP analysis is essential for publication value.

Good Luck at curAHack 2026!

You are working on a real, unsolved problem.

Even incremental improvements to microbiome-based CVD prediction could contribute to non-invasive screening tools that save lives.

Key Resources:

Codes: https://git.opendfki.de/yiwashita/curahack-2026-challenge-5 | Qiita Study: 10317 (AGP) | Paper DOI: 10.1038/s41598-024-82418-3